Assignment # 1

Human Computer Interaction and Computer Graphics

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Abdullah Tallat

BSSE012315010

Object: Toothbrush

Original Toothbrush (standard manual):

Why this matters:

- Simple and widely used
- Easy to understand what it does
- Cheap to produce

Good: easy to hold and use
Bad: hard to reach back teeth areas and many people brush incorrectly → plaque remains.

Alternative Design #1 – Electric Toothbrush:

An electric toothbrush image.

Why it's good:
Rotating/oscillating brush cleans better with less effort
Timer helps users brush longer

Why it's bad:
More expensive
Needs charging / batteries
Bigger and heavier than manual, not good for everyone

Alternative Design #2 – Eco-Friendly/Sustainable Toothbrush:

Bamboo toothbrush images.

Why it's good:
Uses sustainable material → less plastic waste
Eco-friendly appeal

Why it's bad:
Bristles may wear faster
Not as durable for long trips
Some users find it less comfortable

Alternative Design #3 – Smart Toothpaste-Tube-Inspired Toothbrush

Creative toothbrush redesign or toothpaste tube redesign (clever tube)

Example concept:
Smart toothpaste tube that squeezes out all paste and stands upright easily. This inspires design thinking for a toothbrush that maybe collapses or reshapes for travel.

Why it's good:
Encourages less waste & user convenience
Shows design thinking & usability improvement

Why it's bad:
Might be too complex or expensive
Could confuse users if very unconventional

1. **Original Toothbrush** – simplest, familiar, but limited in cleaning effectiveness.
2. **Electric Toothbrush** – improved functionality, better cleaning, but higher cost & power needs.
3. **Eco-Friendly Toothbrush** – better for environment, but usability trade-offs.
4. **Innovative Concept** – shows future possibilities but may not yet be practical.

Picture #1 – Traditional Analog Twin Bell Alarm Clock:

Example: Classic metal bell alarm clock.

Good Design

- Loud bell → very effective wake-up feedback
- Simple mechanism (wind-up)
- No electricity required
- Strong visibility (large hands)

Bad Design

- Hard to set exact alarm time (small dial at back)
- No snooze button
- Can be confusing for children
- No night visibility (unless light is on)

HCI Issue: Poor precision & low user control.

Picture #2 – Digital Alarm Clock (LED Display):

Good Design

- Clear numeric display → high visibility
- Easy to set time with buttons
- Usually includes snooze
- Shows AM/PM clearly

Bad Design

- Too many buttons can confuse users
- Bright LED light may disturb sleep
- Small labels on buttons (poor affordance)

HCI Principle:
Good feedback but sometimes poor learnability.

Picture #3 – Projection Alarm Clock

Good Design

- Projects time on ceiling → no need to move head
- Very convenient at night
- Modern & attractive design

Bad Design

- Expensive
- Projection may blur
- Needs electricity
- Complex setup

HCI Note:
Good visibility, but increased complexity reduces usability.

Picture #4 – Puzzle / Runaway Alarm Clock (Creative Design):

Good Design

- Forces user to get out of bed
- Creative solution for heavy sleepers
- Fun & engaging

Bad Design

- Can fall and break
- Too noisy or stressful
- Not suitable for shared rooms

HCI Concept:
Innovative interaction design — but may reduce user satisfaction.